

BioCoLoop: PUBLIC-HARNESS COMPARISON

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Public frameworks use their task-adapted native experimental controllers and the common biological fitting interface. All Panel-B controllers use Qwen2.5-7B-Instruct, ten laboratories, the same twelve-design library and at most six candidate proposals. Scored cells pass independent recomputation from saved predictions; F_{tool} records a terminal controller-tool protocol failure rather than a predictive score.

Model / method	DTI	Proteomics	Cell perturbation		
	BindingDB	PTPC	VCC	Norman	Tahoe
	AUROC $\uparrow$	AP $\uparrow$	Top-1 $\uparrow$	Top-1 $\uparrow$	Top-1 $\uparrow$
A. Core comparison, seeds 42–44 (mean $\pm$ SD)					
Fixed model (1 lab)	82.83 $\pm$ 1.22	25.05 $\pm$ 4.56	29.94 $\pm$ 0.00	25.56 $\pm$ 0.00	19.40 $\pm$ 0.00
Qwen direct (1 lab)	89.89 $\pm$ 1.02	20.13 $\pm$ 0.07	21.09 $\pm$ 4.38	14.15 $\pm$ 0.00	15.42 $\pm$ 6.21
BioCoLoop (10 labs)	93.76 $\pm$ 1.11	36.99 $\pm$ 0.22	31.54 $\pm$ 0.08	<u>22.25</u> $\pm$ 0.45	22.39 $\pm$ 2.59
B. Matched research-controller comparison, seed 42					
Qwen direct (1 lab)	88.82	<u>20.21</u>	<u>21.73</u>	<u>14.15</u>	<u>13.43</u>
AI-Scientist-v2 (10 labs)	95.07	37.12	31.63	22.74	23.88
AI-Researcher (10 labs)	F_{tool}	37.12	31.63	22.74	23.88
BioCoLoop (10 labs)	<u>94.60</u>	37.12	31.63	22.74	23.88

Table 1: Main results. A: prespecified three-seed core mean $\pm$ SD. B: matched seed-42 task-adapted controllers with 10 laboratories, 12 designs and ≤ 6 candidates. All research controllers in B use Qwen2.5-7B-Instruct. Scores are $\times 100$; bold/underline mark best/second-best within each panel; F_{tool} denotes a controller tool-transport failure.

Outcome. BioCoLoop is top or tied on proteomics and all three cell-perturbation endpoints. On DTI, it reaches 94.60% AUROC, 0.47 points below AI-Scientist-v2. AI-Researcher completes and verifies the other four endpoints but fails the registered one-tool-call protocol on DTI after three identical invalid planning responses.